

# Marco Aurelio Costa

Software Engineer - Pipelines & Infrastructure | Eligible for U.S. security clearance

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## Education

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### Chapman University | Orange, CA

*M.S. Computer Science and Electrical Engineering*

August 2024 - December 2025

Coursework: Statistical ML, Advanced AI, Bayesian Data Analysis, High Performance Computing, Database Systems

*B.S. Computer Science, Minor: Information Security & Policy*

August 2022 - May 2024

Coursework: Algorithm Analysis, Database Management, Numerical Analysis, Linear Algebra

## Experience

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### BioLumen - Operations Analyst & Bookkeeper | San Francisco, CA

May 2024 - Present

- Prepared balance sheets, profit-and-loss statements, and supporting financial records for external investors
- Monitored and reported Shopify sales, revenue, customer, and product analytics, highlighting performance trends
- Developed financial projections and visualizations to support operational planning and comparison across business operations
- Coordinated recurring financial disclosures with executives, legal counsel, accountants, and international investors

### Faculty-Sponsored Independent ML Researcher | Chapman University

January 2026 - June 2026

- Conducted independent research with faculty sponsorship and access to Chapman's eight-GPU V100 computing cluster
- Designed an end-to-end research pipeline spanning computer vision, imitation learning, policy modeling, and simulation
- Analyzed behavior cloning model performance and transitioned toward candidate ranking, simulation, PUCT, and MCTS
- Authored and submitted research manuscript describing pipeline to IEEE MLSP 2026 (accepted August 1st)

## Selected Projects

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### Video-to-State Reconstruction Pipeline

January 2026 - June 2026

*Python, Linux, Bash, Jupyter, Ultralytics YOLOv8, PaddleOCR*

- Built end-to-end imitation learning dataset on 500+ hours of downsampled video, generating 200M+ structured observations
- Trained object detection model achieving 0.995 mAP@0.5 on a synthetic dataset of 480K labeled instances across 12K images
- Trained character recognition model achieving 98.4% character accuracy on a synthetic dataset of 200K images
- Benchmarked OCR models and achieved 3.8× throughput improvement with <1% loss in character accuracy
- Ran distributed inference on machine learning cluster using 8 V100 GPUs to process 21M+ frames in under 60 hours

### Browser-Based Event Labeling & Evaluation Tool

April 2026 - June 2026

*TypeScript, React, Zod, C++, WebAssembly, Hyparquet*

- Built a browser-based tool to define, debug, and label temporal events from sparse OCR and object-detection Parquet timelines
- Implemented manual classification and automated testing of classification configuration and metrics reporting
- Implemented WebAssembly + C++ acceleration for a subset of operations for efficiency boosts when classifying large datasets
- Automated export feature selection, reducing total parameters by ~80% for more efficient downstream DL model training

### Bit-Exact C++ Simulator, MCTS/PUCT Search, Learned Ranker

June 2026 - Present

*C++20, CMake, LuaJIT, PyTorch, XGBoost*

- Trained a branched behavior-cloning transformer to 70.4% top-1 / 91% top-3 expert-action accuracy
- Reimplemented the game's mechanics as a bit-exact C++20 simulator cutting per-decision search from minutes to seconds
- Trained an XGBoost candidate ranker on 24K expert demonstrations, pruning the action space 50–85% at 95% recall

## Open Source Contributions

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### [LightZero AlphaZero Search-Tree Memory Fix PR #492](#)

July 2026

*OpenDILab | C++, MCTS*

- Diagnosed C++ ownership defects in AlphaZero and Gumbel AlphaZero search trees and submitted PR using weak\_ptr and unique\_ptr ownership, reducing measured memory growth from 368.3 to 0.4 KB/tree and 242.8 to 4.0 KB/move

## Skills

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- Programming: Python, C++, Java
- Machine Learning: PyTorch, Policy Transformers, Monte Carlo Tree Search
- GPU / Parallel Computing: Multi-GPU Training, Distributed Inference
- Data & Systems: PySpark, Polars, Parquet, Multiprocessing, Data Pipelines
- Systems: Linux, Bash
- Computer Vision / OCR: OpenCV, YOLO, PaddleOCR, TrOCR, Object Tracking